Course Project 2: implementation of the 5-point scheme for the 2D Poisson equation

Introduction and motivation

The 2D Laplace operator and the associated Poisson problem are fundamental in modeling various physical phenomena across multiple disciplines.

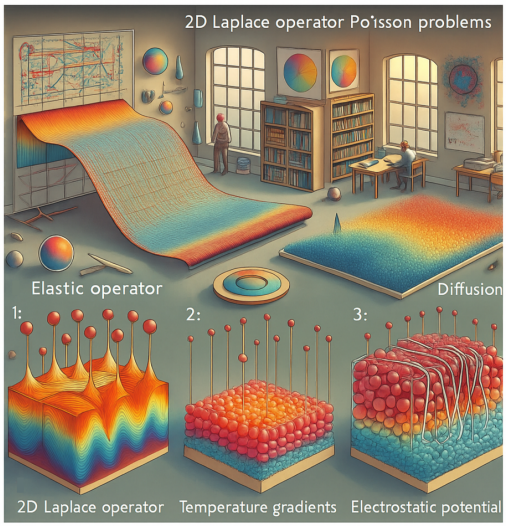

Figure 1: A pictorial illustration of the applications of the Laplace/Poisson operator

In linear elasticity, the Laplace operator describes the deformation of a thin elastic membrane under applied forces, with the Poisson equation governing how external loads affect the membrane's shape at equilibrium. In diffusion processes, it models the steady-state distribution of temperature or concentration of chimical species, where the Poisson problem can describe how a heat or a chemical source influences the equilibrium temperature in a region, such as a room, or the concentration to understand how chemical concentrations reach stable distributions in

biological tissues or in chemical reactors. In biology, the Poisson equation is used to model electrostatic potentials in cells and tissues.
These interpretations highlight the versatility of the 2D Laplace operator in capturing equilibrium states across diverse systems, making it a powerful tool in applied mathematics and physics. This project involves the implementation of the 5-point Finite Difference scheme for solving the Poisson equation on a two-dimensional domain. Students will explore the discretisation process, perform an error analysis, and implement the scheme in `MATLAB` to solve a specified boundary value problem. Additionally, a detailed discussion of the implementation and results is expected.

The Laplace operator in two dimensions

The Laplace operator in two dimensions is defined as:

$$\Delta u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}.$$

Our goal is to solve a Poisson equation of the form:

$$-\Delta u = f \quad \text{in } \Omega,$$

where $\Omega = (0, L_x) \times (0, L_y)$ is a rectangular domain.

Grid definition and discretization

The computational grid for the finite difference method is obtained by dividing the domain into a set of points. Given two integers N_x and N_y, we define:

$$x_i = i \cdot h_x, \quad h_x = \frac{L_x}{N_x}, \quad i = 0, \dots, N_x,$$

$$y_j = j \cdot h_y, \quad h_y = \frac{L_y}{N_y}, \quad j = 0, \dots, N_y.$$

The nodes (x_i, y_j) are the locations where we approximate the solution u.

The 5-Point finite difference scheme

In the 5-point scheme, we approximate the second derivatives as follows:

$$\frac{\partial^2 u}{\partial x^2}(x_i, y_j) \approx \frac{u(x_{i+1}, y_j) - 2u(x_i, y_j) + u(x_{i-1}, y_j)}{h_x^2},$$

$$\frac{\partial^2 u}{\partial y^2}(x_i, y_j) \approx \frac{u(x_i, y_{j+1}) - 2u(x_i, y_j) + u(x_i, y_{j-1})}{h_y^2}.$$

Combining these yields the discrete approximation of the Laplace operator:

$$\Delta u(x_i, y_j) \approx \frac{1}{h_x^2}\left(u_{i+1,j} - 2u_{i,j} + u_{i-1,j}\right) + \frac{1}{h_y^2}\left(u_{i,j+1} - 2u_{i,j} + u_{i,j-1}\right).$$

Note that, under the assumption $h_x = h_y = h$, it holds:

$$\Delta u(x_i, y_j) \approx \frac{1}{h^2}\left(u_{i+1,j} + u_{i,j+1} - 4u_{i,j} + u_{i-1,j} + u_{i,j-1}\right).$$

Matrix formulation of the discretized problem

The finite difference approximation of the Poisson problem can be written as a system of equations:

$$-\frac{1}{h^2}\left(u_{i+1,j} + u_{i,j+1} - 4u_{i,j} + u_{i-1,j} + u_{i,j-1}\right) = f(x_i, y_j),$$

for $i = 1, \ldots, N_x - 1$ and $j = 1, \ldots, N_y - 1$. Note that, for the sake of simplicity, we used the assumption $h_x = h_y = h$ for writing the formula above.
The system can be represented in matrix form as:

$$A\boldsymbol{U} = \boldsymbol{F},$$

where A is the coefficient matrix, $\boldsymbol{U}$ is the vector of unknowns, and $\boldsymbol{F}$ is the vector representing the right-hand side terms, including boundary conditions.

Properties of the 5-point scheme

The 5-point finite difference scheme for the discretization of the 2D Laplace operator has several notable properties, making it a popular choice for solving partial differential equations in two dimensions.

- **Consistency**: the scheme is consistent with the continuous Poisson problem and provides

a second-order accurate approximation. This means that as the grid spacing h tends to zero, the local truncation error goes to zero at a rate proportional to h^2.

- **Stability** (in space): due to the centered nature of the Laplace operator, the 5-point scheme is stable, implying that small perturbations in the input (such as rounding errors) do not lead to unbounded growth in the solution. This stability is crucial for ensuring that numerical solutions remain bounded and reliable over repeated applications.

- **Sparsity-preserving**: the resulting matrix A in the discretized system is sparse. In the case of Cartesian grid with suitable oredered numbering of the nodes, it has non-zero entries only near the diagonal, speci fically forming a 5-point stencil. This sparsity pattern significantly reduces memory requirements and allows for efficient computational methods, such as iterative solvers, to be applied effectively.

- **Symmetry and positive definiteness**: the matrix A generated by the 5-point scheme is symmetric and positive definite. Symmetry ensures that the solution process is simplified and can leverage algorithms optimized for symmetric matrices. Positive definiteness guarantees the existence of a unique solution for the discretized system, making the scheme particularly reliable for solving elliptic PDEs involving the Laplace operator, such as the Poisson problem.

These properties make the 5-point Finite Difference scheme a robust and efficient method for numerically solving the Laplace equation on a 2D Cartesian grid.

Assembly of the matrix A

To solve the discretized 2D Poisson equation, we need to assemble the matrix A, which represents the discrete Laplace operator. The assembly of A involves mapping the 2D grid points $(\boldsymbol{x}_i, \boldsymbol{y}_j)$ by means of a single index k in a one-dimensional vector $\boldsymbol{U}$ containing all unknowns $u_{ij} \simeq u(\boldsymbol{x}_i, \boldsymbol{y}_j)$. In this section, to keep the introduction as general as possible, we consider the parameters h_x and h_y to be possibly different.

Mapping grid points to vector indices

Each double index (i, j) is mapped to an integer k for easier manipulation in matrix-vector operations. This mapping can be achieved by ordering the nodes in two ways: by rows or by columns. The mapping schemes are as follows:

- **Ordering by rows**: $U_k = u_{ij}$ with $k = (j-1)\,N_x + i$. reads by rows, then stacks

- **Ordering by columns**: $U_k = u_{ij}$ with $k = (i-1)\,N_y + j$. reads by colums, then stacks

For this assembly, we adopt the row ordering. Switching i with j and N_x with N_y would yield the corresponding matrix formulation for column ordering.

Matrix representation of the discretised Laplace operator

We aim to rewrite the finite difference scheme for the Laplace operator as follows:

$$A\boldsymbol{U} = \boldsymbol{F}, \quad A \in \mathbb{R}^{N\times N}, \quad \boldsymbol{U} \in \mathbb{R}^{N}, \quad \boldsymbol{F} \in \mathbb{R}^{N},$$

where A and $\boldsymbol{F}$ are defined such that their coefficients depend on $1/h_x^2$ and $1/h_y^2$. Here, $N = N_x \times N_y$ represents the total number of grid points.

Block structure of A

To assemble A, we proceed by blocks. We define $D \in \mathbb{R}^{N_x\times N_x}$ as the block that handles coupling within the same row, while $E \in \mathbb{R}^{N_x\times N_x}$ accounts for coupling between adjacent rows. The constants in the matrix assembly are given by:

$$\alpha = \frac{2}{h_x^2} + \frac{2}{h_y^2}, \quad \beta = -\frac{1}{h_x^2}, \quad \gamma = -\frac{1}{h_y^2}.$$

The blocks D and E are constructed as follows:

$$D = \begin{bmatrix} \alpha & \beta & 0 & \dots & 0 \\ \beta & \alpha & \beta & \dots & 0 \\ 0 & \beta & \alpha & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \beta \\ 0 & \dots & 0 & \beta & \alpha \end{bmatrix}, \quad E = \begin{bmatrix} \gamma & 0 & \dots & 0 \\ 0 & \gamma & \dots & 0 \\ 0 & 0 & \ddots & 0 \\ 0 & \dots & 0 & \gamma \end{bmatrix}.$$

Constructing the global matrix A

By arranging the blocks D and E appropriately, we construct A as a block tridiagonal matrix:

$$A = \begin{bmatrix} D & E & 0 & \dots & 0 \\ E & D & E & \dots & 0 \\ 0 & E & D & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & E \\ 0 & \dots & 0 & E & D \end{bmatrix}.$$

Assembly of the right-hand side vector $\boldsymbol{F}$

Analogous to the block decomposition of A, we proceed with the decomposition of $\boldsymbol{F}$ into sub-vectors $\boldsymbol{F}_j \in \mathbb{R}^{N_x}$, each corresponding to the source terms and boundary values on the j-th row of the grid. After arranging all blocks, the final global form of $\boldsymbol{F}$ is:

$$\boldsymbol{F} = \Big[\boldsymbol{F}_1,\ \boldsymbol{F}_2,\ \ldots,\ \boldsymbol{F}_{Ny}\Big]^T.$$

Please notice that in the aformentioned source terms, the boundary conditions have not been still taken into account, so this formulation for $\boldsymbol{F}$ is very general. Be aware to suitably derive it for the problem at hand during the project.
With the matrix A and vector $\boldsymbol{F}$ fully assembled, we have a complete linear system representation of the finite difference scheme:

$$A\boldsymbol{U} = \boldsymbol{F}.$$

Solving this system provides an approximation to the solution $u(x, y)$ over the domain Ω.

Project tasks

Now suppose we consider a rectangular sample of a biological tissue (e.g., cartilage, arterial wall) that can be represented by an elastic membrane. The elastic properties of the material are represented by an accurate choice of the elastic parameter μ. By imposing a force f on the test piece and fixing the boundary conditions g at the boundary, we can observe the deformation of the test piece as a function of the stress considered. This experiment can be very useful for characterizing stress testing of biological samples.

This project consists of the following main tasks:

1. **Error Analysis**: perform an error analysis of the 5-point finite difference scheme. Following the methodology of the 1D case, assume that the scheme is stable and compute the error order of accuracy for the 2D case. Discuss the findings and theoretical implications.

2. **MATLAB Implementation**: implement the 5-point finite difference scheme in `MATLAB` to numerically solve the problem outlined above. Write a function that computes the approximate solution u on a grid covering the domain $\Omega = (0, 1)^2$.

3. **Numerical error analysis**: Use $\mu = 2$, $f = -8$ and Dirichlet boundary conditions $g = x^2 + y^2$ on $\partial\Omega$. Approximate the analytical solution $u(x, y) = x^2 + y^2$ of the Poisson problem. Then, solve the problem via successive halvings of h_x and h_y, compute the errors as $E = \max_i \max_j |u_h(x_i, y_j) - u_{i,j}|$ and plot the errors versus h in *log-log scale*. Comment on the results. (*Hint*: for the sake of simplicity, consider $h = h_x = h_y$).

4. **Convergence test**: Use a problem of your choice to numerically prove the second-order of convergence of the implemented method (*Hint*: you can change also the exact solution!).

5. **Discussion of Code and Results**: provide a concise discussion of the `MATLAB` code, emphasizing the main ideas and any notable implementation details without including the full code in this document. Analyze the numerical results and discuss any challenges faced during implementation. Highlight how the observed results align with the theoretical expectations and any deviations.

6. **Extra - Neumann boundary conditions** Approximate the following problem Poisson problem:
$$\begin{cases} -\Delta u = 0 & \text{in } \Omega \\ \partial_\nu u = \sin(2\pi s) & \text{on } \partial\Omega \end{cases}$$
where $\Omega = (0,1)^2$, $\partial_\nu u$ denotes the normal derivative to the boundary and s denotes the length of the arc along the boundary $\partial\Omega$, originating at $(0,0)$. To guarantee well-posedness of the problem, we consider the solution to have a zero average on Ω. Is the solution a harmonic function?

Evaluation criteria

The laboratory project consists in answering the questions reported in the document uploaded on WeBeep.

The delivery of the project and the oral presentation are intended to be individual.

During the oral session of the corresponding written exam, the students that have decided to take the laboratory project should expose the activity through some slides (or in general a written document), highlighting the implementation issues (not reporting in the slides or document all the code, just the main ideas) and a discussion of the results. The slides should be prepared for a presentation of 10 minutes. The teacher could ask the student to open and run the implemented code and in real time to change something in order to evaluate the knowledge and the degree of learning of the student on the topic.

The oral exam (as well as the standard oral one on the theoretical topics) will allow the student to have an increment between -3 and $+3$ points with respect to the mark obtained at the written exam.